

ADNAN MASHRUR SADAD

FULL STACK SOFTWARE ENGINEER | REACT AND PYTHON

Kuala Lumpur, Malaysia | +60 11-3968 7435 | adnanmashrursadad@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Universiti Teknologi Malaysia (UTM), MJIIT

Kuala Lumpur, Malaysia

B.Sc. Computer Science (Software Engineering) with Honours — CGPA 3.43/4.00

2021 – 2026

- Dean's List 2022/2023 and 2025/2026. Thesis: OCR-based personal finance tracker performing automated, income-tax-ready expense classification.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C++, Dart

Frontend: React, Next.js, HTML/CSS, server-sent events, responsive UI

Backend: FastAPI, SQLAlchemy, Pydantic, Alembic, REST APIs, OpenAPI, SQLite, microservices

Delivery: Git, Docker, GitHub Actions, CI/CD, pytest, Playwright, unit and integration testing, Agile/Scrum, full SDLC

EXPERIENCE

Software Developer Intern (Hybrid)

Oct 2024 – Mar 2025

Joget Inc. — enterprise workflow automation on the Joget DX low-code platform

Kuala Lumpur, Malaysia

- Shipped workflow automations and UI components used by client teams across [N] departments, owning the full SDLC in Agile sprints alongside senior engineers, QA and product managers.
- Removed manual re-entry steps from recurring client workflows by integrating REST APIs with external services.
- Maintained and released modules through CI/CD across [N] recurring production release cycles for enterprise clients.

PROJECTS

InterviewPilot — AI Mock Interview Platform

2026

React + TypeScript, FastAPI, SQLAlchemy, Groq API, Docker, GitHub Actions, pytest | [GitHub Link](#)

- Built an end-to-end interview platform on React/TypeScript and FastAPI with persistent sessions, resumable interviews, idempotent answer submission and shareable reports backed by SQLAlchemy.
- Kept the product usable when the AI provider fails, by bounding follow-up question depth, validating every model response against a schema with a re-prompt repair loop, and explicitly labelling fallback scores rather than degrading silently.
- Shipped behind 89% statement coverage across 19 backend tests plus production builds and a browser flow covering session creation, answer submission and report generation.

FinScout — Streaming Company-Research Application

2026

Next.js, FastAPI, server-sent events, FAISS | [GitHub Link](#)

- Built a Next.js and FastAPI interface for company research and Q&A that streams tool progress and report generation to the browser over server-sent events, backed by filing retrieval and FAISS search.
- Hardened the streaming path against the failure modes it actually hits — blank requests, upstream provider errors and interrupted streams — covered by 52 backend tests, three stream tests and fixture-driven browser checks on report and error rendering.

Agentic Planner — Typed Workflow Execution Layer

2026

Python, dependency validation, provenance checks, deterministic testing | [GitHub Link](#)

- Closed a class of silent-failure bugs in tool execution by enforcing argument provenance, dependency checks and confirmation gates, verified by 84 offline tests and four deterministic workflow scenarios.

RESEARCH

Deceptive Grounding in Regulatory RAG (SAFE-RAG)

2025 – 2026

First author — short paper under review at ALTA 2026 | [GitHub Link](#)

- Measured an 8.6% rate (95% CI 4.5–13.5%) of schema-valid, citation-entailed RAG answers that are nonetheless wrong, across 200 annotated items from a financial-regulation corpus (Cohen's $\kappa = 0.786$).

Answerability under Schema Evolution in Text-to-SQL

2026

First author — full manuscript complete (v1.0); not yet submitted

- Built a 1,382-instance benchmark over 20 Spider databases and ran 27,384 generations on free-tier Kaggle T4 GPUs; shipped a detector recovering 79.5% of structurally invisible schema drift at a 13.0% false-alarm rate.

AWARDS & LEADERSHIP

- **Gold Medal and Best Video Award**, myHCI-UX Student Design Challenge (Oct 2025) — national inter-university UX design competition.
- Secretary Treasurer, SOFEA Society (2022–2023); organised AI workshops for 200+ students.